

Equalization: current implications and structural observations

Research notes, 30 September 2026

Proposition 1 (An improved known relation between the optima). *Let A_n be the shortest arbitrary-composition permutation-fair string length, and let E_n be the corresponding equal-composition optimum. The existing Hahn construction and exponential lower bound already imply*

$$A_n \leq E_n \leq A_n^{O(\log n)} \leq A_n^{O(\log \log A_n)}.$$

This is stronger than the relation $E_n \leq A_n^{O(\log A_n)}$ obtained by independent coordinate blowups alone.

Proof. For absolute positive constants c, C and all sufficiently large n , $\log A_n \geq cn$ and $\log E_n \leq Cn \log n$. Consequently $\log E_n \leq (C/c) \log A_n \log n$. Finally $n = O(\log A_n)$. $\square$

Proposition 2 (A sharper independent-blowup bound). *For an n -fair string of total length A and counts $c_1, \dots, c_n$, one can equalize to length*

$$n \operatorname{lcm}(c_1, \dots, c_n) \leq \frac{n \prod_i c_i}{\prod_{p \leq n} p^{n-p}} \leq \frac{n(A/n)^n}{\prod_{p \leq n} p^{n-p}}.$$

Proof. Blow up letter i by $\operatorname{lcm}(c_1, \dots, c_n)/c_i$. At least $n - p + 1$ counts are divisible by each prime $p \leq n$. Therefore

$$v_p \left(\frac{\prod_i c_i}{\operatorname{lcm}(c_1, \dots, c_n)} \right) = \sum_i v_p(c_i) - \max_i v_p(c_i) \geq n - p.$$

Multiply these estimates and apply AM–GM. $\square$

Lemma 3 (Concatenation with proportional compositions). *Let $s_1, \dots, s_B$ be permutation-fair strings on the same alphabet, and suppose their multiplicity vectors are $a_b c$ for a fixed vector $c \in \mathbb{R}_{>0}^n$ and positive scalars a_b . Then $s_1 \cdots s_B$ is permutation-fair, with vector $(\sum_b a_b) c$.*

Proof. Fix a pattern π on a set S of distinct letters and let q_b be the number of its consecutive letters selected from block b . Partial fairness gives contribution

$$\left(\prod_{i \in S} c_i \right) \prod_{b=1}^B \frac{a_b^{q_b}}{q_b!}.$$

Summing over $q_1 + \dots + q_B = |S|$ gives

$$C_\pi(s_1 \cdots s_B) = \frac{(\sum_b a_b)^{|S|}}{|S|!} \prod_{i \in S} c_i,$$

independently of the order of π . $\square$